Geometric Formulation of 4-Dimensional Spacetime via Gnomonic Projection

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Abstract

This note examines, from a purely geometric viewpoint, the deformation of the induced metric tensor that arises when 4-dimensional Euclidean space is mapped onto the surface of a 5-dimensional hypersphere of curvature radius R via the gnomonic (central) projection. In particular, we present step-by-step derivations of: (i) the exact induced metric tensor $g_{\mu\nu}$ (including the contribution of the fifth component), and (ii) the geometric relationship between the Einstein tensor and the constant $\Lambda = 3/R^2$. Physical interpretation is deferred to future work; no cosmological assumptions are made in this note.

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1 Introduction

The gnomonic projection is defined as the central projection from the center of a sphere onto a tangent plane, and its most important property — that great circles (geodesics) on the sphere map to straight lines on the plane — is well established [1].

This note extends this projection from 4 to 5 dimensions and investigates the metric structure that arises in the resulting coordinate system. The problem is purely geometric; physical interpretations (gravity, spacetime, relativity, etc.) lie outside the scope of this note.

2 Preliminaries: Definition and Basic Properties

2.1 Gnomonic Projection in General Dimension

Definition 2.1 (n-dimensional gnomonic projection [1, 2])**.** *In n-dimensional Euclidean space, consider a tangent hyperplane Π_R at distance R from the center of a sphere (with coordinates $(x^1, \dots, x^n) \in \mathbb{R}^n$, origin at the point of tangency). Setting $l = \sqrt{R^2 + \sum_{i=1}^n (x^i)^2}$, we call*

$$\Phi : \Pi_R \to S^n(R) \subset \mathbb{R}^{n+1}, \quad (x^1, \dots, x^n) \mapsto \left(\frac{Rx^1}{l}, \dots, \frac{Rx^n}{l}, \frac{R^2}{l} \right) \tag{1}$$

the gnomonic projection*, where $S^n(R)$ is the n-sphere of radius R embedded in $\mathbb{R}^{n+1}$:*

$$S^n(R) = \left\{ (Y^1, \dots, Y^{n+1}) \in \mathbb{R}^{n+1} \;\middle|\; \sum_{A=1}^{n+1} (Y^A)^2 = R^2 \right\}.$$

Proposition 2.2 (Geodesic preservation [1])**.** *Under the gnomonic projection, any straight line on the tangent plane maps to a great circle (geodesic) on the sphere.*

3 Derivation of the Metric Tensor

3.1 Setup

We set $n = 4$ and consider the gnomonic projection from the tangent hyperplane Π_R (coordinates (x^1, x^2, x^3, x^4)) to $S^4(R) \subset \mathbb{R}^5$. Since the tangent plane is Euclidean, $x_\mu = x^\mu$.

By Definition 2.1, the 5-dimensional target coordinates are

$$(Y^1, Y^2, Y^3, Y^4, Y^5) = \left(\frac{Rx^1}{l}, \frac{Rx^2}{l}, \frac{Rx^3}{l}, \frac{Rx^4}{l}, \frac{R^2}{l}\right), \tag{2}$$

$$l = \sqrt{R^2 + \textstyle\sum_{\mu=1}^4 (x^\mu)^2}, \quad r^2 = \textstyle\sum_{\mu=1}^4 (x^\mu)^2, \quad l^2 = R^2 + r^2. \tag{3}$$

3.2 Derivation of the Pullback Metric

The induced metric on $S^4(R) \subset \mathbb{R}^5$ is determined by the standard inner product on $\mathbb{R}^5$ via [6]:

$$dS^2 = \sum_{A=1}^{5} (dY^A)^2. \tag{4}$$

We compute all five components, including the fifth.

Components 1–4 ($Y^\mu = Rx^\mu/l$):

Lemma 3.1.

$$\frac{\partial Y^\mu}{\partial x^\nu} = \frac{R}{l}\left(\delta^\mu_\nu - \frac{x^\mu x^\nu}{l^2}\right). \tag{5}$$

Proof. $\dfrac{\partial}{\partial x^\nu}\left(\dfrac{Rx^\mu}{l}\right) = \dfrac{R\delta^\mu_\nu}{l} + Rx^\mu\left(-\dfrac{x^\nu}{l^3}\right) = \dfrac{R}{l}\left(\delta^\mu_\nu - \dfrac{x^\mu x^\nu}{l^2}\right).$ □

Therefore,

$$\sum_{\mu=1}^{4} (dY^\mu)^2 = \frac{R^2}{l^2} \sum_{\nu,\rho=1}^{4} \left(\delta_{\nu\rho} - \frac{2x_\nu x_\rho}{l^2} + \frac{r^2 x_\nu x_\rho}{l^4}\right) dx^\nu dx^\rho. \tag{6}$$

Fifth component ($Y^5 = R^2/l$):

$$\frac{\partial Y^5}{\partial x^\nu} = -\frac{R^2 x^\nu}{l^3}, \qquad \left(dY^5\right)^2 = \frac{R^4}{l^6}\, x_\nu x_\rho\, dx^\nu dx^\rho. \tag{7}$$

Summing all components. The coefficient of $x_\nu x_\rho$ from (6) and (7) is

$$-\frac{2R^2}{l^4} + \frac{R^2 r^2}{l^6} + \frac{R^4}{l^6} = -\frac{2R^2}{l^4} + \frac{R^2(r^2 + R^2)}{l^6} = -\frac{2R^2}{l^4} + \frac{R^2}{l^4} = -\frac{R^2}{l^4}.$$

Therefore,

$$\boxed{g_{\mu\nu} = \frac{R^2}{l^2}\left(\delta_{\mu\nu} - \frac{x_\mu x_\nu}{l^2}\right).} \tag{8}$$

Remark (Indispensability of the fifth component)**.** *If the fifth component $(dY^5)^2$ is omitted, the coefficient of $x_\nu x_\rho$ remains $-2R^2/l^4 + R^2 r^2/l^6$ and (8) cannot be obtained. The contribution $+R^4/l^6 = +R^2/l^4 - R^2 r^2/l^6$ from the fifth component exactly compensates this deficit.*

4 Christoffel Symbols and Riemann Curvature

Theorem 4.1 (Curvature tensor of a space of constant curvature [3–5])**.** *The Riemann curvature tensor of the n-sphere $S^n(R)$ is*

$$R_{\mu\nu\rho\sigma} = \frac{1}{R^2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}). \tag{9}$$

Remark. *Equation (9) is the standard result for spaces of constant curvature [4, 5]. The direct derivation in gnomonic coordinates from the Christoffel formula is planned for Appendix A; here we cite [3, 4].*

5 Ricci Tensor and Einstein Tensor

5.1 Ricci Tensor

Proposition 5.1. *The Ricci tensor of the n-sphere $S^n(R)$ is*

$$R_{\mu\nu} = \frac{n-1}{R^2}g_{\mu\nu}. \tag{10}$$

Proof. Substituting (9) into $R_{\mu\nu} = g^{\rho\sigma}R_{\mu\rho\nu\sigma}$:

$$g^{\rho\sigma}R_{\mu\rho\nu\sigma} = \frac{1}{R^2}g^{\rho\sigma}(g_{\mu\nu}g_{\rho\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

First term: $g^{\rho\sigma}g_{\mu\nu}g_{\rho\sigma} = n\,g_{\mu\nu}$. Second term: $g^{\rho\sigma}g_{\mu\sigma}g_{\nu\rho} = g_{\nu\mu} = g_{\mu\nu}$. Hence $R_{\mu\nu} = \frac{1}{R^2}(n-1)g_{\mu\nu}$. □

5.2 Ricci Scalar

$$R_{\text{scalar}} = g^{\mu\nu}R_{\mu\nu} = \frac{n(n-1)}{R^2}. \tag{11}$$

For $n = 4$: $R_{\text{scalar}} = 12/R^2$.

5.3 Einstein Tensor

From the definition [4]:

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R_{\text{scalar}}. \tag{12}$$

For $n = 4$:

$$G_{\mu\nu} = \frac{3}{R^2}g_{\mu\nu} - \frac{6}{R^2}g_{\mu\nu} = -\frac{3}{R^2}g_{\mu\nu}. \tag{13}$$

6 Main Theorem

Theorem 6.1 (Main result)**.** *When the tangent hyperplane Π_R of 4-dimensional Euclidean space is mapped to $S^4(R)$ via the gnomonic projection, the induced metric $g_{\mu\nu}$ (eq. (8)) satisfies*

$$G_{\mu\nu} + \Lambda\, g_{\mu\nu} = 0 \qquad \textit{with} \qquad \boxed{\Lambda = \frac{3}{R^2}.} \tag{14}$$

Proof. From (13), $G_{\mu\nu} = -\frac{3}{R^2} g_{\mu\nu}$; setting $\Lambda = \frac{3}{R^2}$ gives (14) immediately. □

Remark. *Equation (14) is a* geometric identity *for $S^4(R)$ with Euclidean signature and is not to be interpreted as a physical law. It is structurally distinct from de Sitter spacetime with Lorentzian signature; the two are related by analytic continuation (Wick rotation $t \to it$, see [5] Appendix D).*

Both the Euclidean and Lorentzian versions share $\Lambda = 3/R^2$, consistent with the Gibbons–Hawking Euclidean quantum gravity framework [8]. The sign difference is a technical issue bridged by Wick rotation.

Remark (General dimension)**.** *For the n-sphere $S^n(R)$, the value of Λ satisfying $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ is*

$$\Lambda_n = \frac{(n-1)(n-2)}{2R^2}$$

(for $n = 4$: $\Lambda_4 = 3/R^2$). The case $n = 4$ is physically distinguished by its correspondence to the de Sitter vacuum solution.

7 Topics for Future Investigation

The following are planned for future work:

- **Geometric interpretation of Wick rotation:** How to describe $t \to it$ within the gnomonic framework.
- **Lorentzian gnomonic projection:** Derivation of the induced metric for $l^2 = R^2 + x^2 + y^2 + z^2 - t^2$.
- **Geodesics under symmetry:** Geodesic equations for $X^2 = X^3 = X^4 = b$, $X^1 = t$.
- **Introduction of matter:** Extension to $T_{\mu\nu} \neq 0$.
- **Euclidean/Lorentzian correspondence:** Proof of the correspondence between unified formula (D.3) and the Beltrami–de Sitter metric of Guo et al. [9–11].

8 Summary of Results

Item	Result	Eq.	Source
Map definition	$Y^\mu = Rx^\mu/l$, $Y^5 = R^2/l$	(2)	[1, 2]
Induced metric	$g_{\mu\nu} = \frac{R^2}{l^2}\left(\delta_{\mu\nu} - \frac{x_\mu x_\nu}{l^2}\right)$	(8)	This work
Riemann curvature	$\frac{1}{R^2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$	(9)	[3–5]
Ricci tensor	$R_{\mu\nu} = \frac{3}{R^2} g_{\mu\nu}$	(10)	This work
Ricci scalar	$R_{\text{scalar}} = \frac{12}{R^2}$	(11)	This work
Einstein tensor	$G_{\mu\nu} = -\frac{3}{R^2} g_{\mu\nu}$	(13)	This work
Main theorem	$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$, $\Lambda = \frac{3}{R^2}$	(14)	This work

Table 1: Summary of main results.

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A Direct Computation of Christoffel Symbols (Planned)

A complete derivation of $\Gamma^{\lambda}_{\mu\nu}$ from the metric (8) via the Christoffel formula, leading directly to the Riemann curvature tensor of Theorem 4.1, is planned for a future report. At present we cite [3,4].

B Flat Limit $R \to \infty$

As $R \to \infty$, $l \to R$, so $g_{\mu\nu} \to \delta_{\mu\nu}$ and $\Lambda = 3/R^2 \to 0$. The metric converges to flat Euclidean metric and the curvature vanishes, corresponding to the gnomonic projection degenerating to the identity map.

C Regularity of the Projection

Proposition C.1 (Regularity)**.** *Let $\Phi : \mathbb{R}^4 \to S^4(R)$ be the gnomonic projection. The following hold:*

1. *$R \to 0$ ($r > 0$ fixed): $Y^5, Y^\mu \to 0$. Image degenerates to the origin but does not diverge.*

2. *$R \to \infty$ (r fixed): $Y^5 \to \infty$; $Y^\mu \to x^\mu$ (finite).*

3. *Fixed $R > 0$: $l \geq R > 0$, so Φ is C^∞ on all of $\mathbb{R}^4$.*

Proof. Direct computation of $l = \sqrt{R^2 + r^2}$: (1) $Y^5 = R^2/l \leq R^2/r \to 0$. (2) $Y^5 = R(1 + r^2/R^2)^{-1/2} \to \infty$. (3) $l \geq R > 0$ for each component. □

D Squared Metric: Unified Description of Euclidean and Lorentzian Geometries

Within the geometric description of the gnomonic projection, the transition from Euclidean to Lorentzian geometry can be described as a choice of the real parameter ϵ_μ.

Remark. *The correspondence between the Lorentzian gnomonic projection ($\epsilon_4 = -1$) and the Beltrami coordinate system of de Sitter spacetime has been established as prior work by Guo et al., starting from the Lorentzian side [9–11]. The originality of this appendix lies in the explicit derivation of the unified formula (D.3), which encompasses both the Euclidean case ($S^4(R)$, [8]) and the Lorentzian case (de Sitter–Beltrami, [9–11]) as special cases of a single ϵ_μ parametrization.*

Observation (Quadratic nature of physical metrics)**.** *All metrics appearing in the gnomonic projection involve only t^2, never t itself:*

$$l^2 = R^2 + x^2 + y^2 + z^2 \pm t^2, \quad ds^2 = \pm dt^2 + dx^2 + dy^2 + dz^2.$$

The sign difference is a choice of coefficient $\epsilon_\mu \in \{+1, -1\}$.

Proposition D.1 (Unified Euclidean/Lorentzian description)**.** *Choose $(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4) \in \{+1, -1\}^4$. Define the embedding metric*

$$dS^2_{\text{emb}} = \sum_{\mu=1}^{4} \epsilon_\mu (dY^\mu)^2 + (dY^5)^2, \tag{D.1}$$

the tangent-plane metric $ds^2_{\text{tan}} = \sum_\mu \epsilon_\mu (dx^\mu)^2$, and $l^2 = R^2 + \sum_\mu \epsilon_\mu (x^\mu)^2$. With the same gnomonic map $Y^\mu = Rx^\mu/l$, $Y^5 = R^2/l$, the induced metric is

$$g_{\mu\nu} = \frac{R^2}{l^2}\left(\epsilon_\mu \delta_{\mu\nu} - \frac{\epsilon_\mu \epsilon_\nu x_\mu x_\nu}{l^2}\right). \tag{D.3}$$

(Einstein summation is not used for repeated indices.)

Proof. Setting $r_\epsilon^2 = \sum_\mu \epsilon_\mu (x^\mu)^2 = l^2 - R^2$ and computing $\sum_\mu \epsilon_\mu (dY^\mu)^2 + (dY^5)^2$, the coefficient of $\epsilon_\nu \epsilon_\rho x_\nu x_\rho$ combines as

$$-\frac{2R^2}{l^4} + \frac{R^2 r_\epsilon^2}{l^6} + \frac{R^4}{l^6} = -\frac{2R^2}{l^4} + \frac{R^2 l^2}{l^6} = -\frac{R^2}{l^4},$$

using $r_\epsilon^2 + R^2 = l^2$. □

Remark (Geometric correspondence with Wick rotation)**.** *Within the geometric description of the gnomonic projection, the Wick rotation $t \to it$ corresponds to $\epsilon_4 : +1 \to -1$:*

$$\begin{aligned} \textit{Euclidean}: \quad & \epsilon_\mu = (+1, +1, +1, +1) \;\Rightarrow\; l^2 = R^2 + x^2 + y^2 + z^2 + t^2, \\ \textit{Lorentzian}: \quad & \epsilon_\mu = (+1, +1, +1, -1) \;\Rightarrow\; l^2 = R^2 + x^2 + y^2 + z^2 - t^2. \end{aligned}$$

Both are special cases of the single framework defined by (D.1). Note that the necessity of Wick rotation for path-integral convergence in quantum gravity lies outside the scope of this note.

The image of the map satisfies

$$\sum_{\mu=1}^{4} \epsilon_\mu (Y^\mu)^2 + (Y^5)^2 = R^2, \tag{D.6}$$

which for $\epsilon_\mu = (+1, +1, +1, -1)$ gives the pseudo-sphere $(Y^1)^2 + (Y^2)^2 + (Y^3)^2 - (Y^4)^2 + (Y^5)^2 = R^2$ in $\mathbb{R}^{4,1}$, structurally corresponding to the de Sitter hypersurface. A rigorous proof of this correspondence starting from (D.3) is deferred to future work.

This note aims to organize geometric facts and offers no physical interpretation. All sources are cited in the references.